
INDUSTRIAL TRAINING PROJECT

Virtual Robotics Testing

Catch failures in simulation. Deploy robots faster and safer.

Live demo | 2D twin today | Gazebo / Isaac when installed

The problem

- Hardware trial-and-error is slow, costly, and unsafe.
- Edge cases (blocked aisles, sensor dropouts) slip into production.
- Gazebo, Webots, and Isaac give physics -- not an automated test lab.
- Digital twins today optimize logistics more than robot control.

The platform

- Digital twin of the warehouse + differential-drive robot.
- AI layer: adversarial what-if search, MCMC, grammar maps.
- Bayesian speed tuning and a CBF safety shield.
- Same scenarios run on twin, Gazebo/ROS 2, or Isaac Sim.

Live demo: what if the aisle is blocked?

Open aisle

Baseline controller reaches the goal.

Open [docs/demo/index.html](#) to play the path.

Blocked aisle

No global planner -- timeout / stuck.

Open [docs/demo/index.html](#) to play the path.

Adversarial search beats random

Method	Failure rate
Random crates	30%
MCMC	55%
Adversarial hill-climb	90%

20 trials on the open-aisle map. Adversarial search pushes a crate onto the lane.

+19.9%

Throughput vs default speed

CBF on

Open aisle still succeeds, 0 violations

3 engines

twin / gazebo / isaac --backend

- Bayesian optimization of cruise speed under the CBF shield.
- Domain-randomized Q-learning is the sim-to-sim proxy (friction 0.8 / 1.0 / 1.15).

How to run

TERMINAL

```
$ python3 -m pytest -q
```

```
$ python3 -m virtual_lab demo
```

```
$ python3 -m virtual_lab complete --out docs/evaluation
```

```
$ python3 -m virtual_lab run --backend gazebo --scenario scenarios/open_aisle.json
```

```
$ gz sim simulators/worlds/open_aisle.sdf
```

ASK

Approve the industrial-training MVP

Today: 2D twin, adversarial search, Bayesian tuning, demo + slides.

Next on a workstation: Gazebo/ROS 2, then Isaac on GPU.

Human review stays in the loop -- the platform never deploys itself.